

Urkunde

Andrei Tregubov,

geboren am 31. März 2001

in Nischni Nowgorod (Russische Föderation),

hat die Masterprüfung für den Studiengang

Computational Mathematics

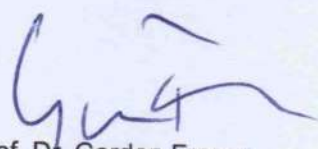
mit der Gesamtnote „gut“ (1,8) bestanden.

Aufgrund dieser Prüfung wird der akademische Grad eines

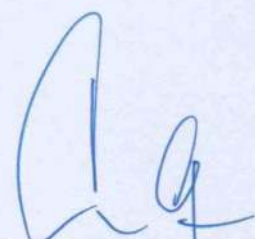
Master of Science (M.Sc.)

verliehen.

Passau, den 1. April 2026.



Prof. Dr. Gordon Fraser
Dekan



Prof. Dr. Thomas Müller-Gronbach
Vorsitzender des Prüfungsausschusses

Zeugnis

Andrei Tregubov,

geboren am 31. März 2001

in Nischni Nowgorod (Russische Föderation),

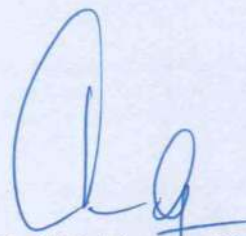
hat die Masterprüfung für den Studiengang

Computational Mathematics

mit der Gesamtnote „gut“ (1,8) bestanden.

Die erbrachten Leistungen befinden sich umseitig.

Passau, den 1. April 2026.



Prof. Dr. Thomas Müller-Gronbach
Vorsitzender des Prüfungsausschusses

Bezeichnung	Note	Anm.	LP
Pflichtbereich			
Seminar 1 zu Mathematik			40,0
Seminar 2 zu Mathematik	2,3		5,0
Präsentation der Masterarbeit	1,0 (a)		5,0
Masterarbeit	1,0 (a)		3,0
	1,3 (a)		27,0
Wahlpflichtbereich			81,0
Modulgruppe Algebra, Geometry and Cryptography			15,0
Cryptography			9,0
Distributed Algorithms	4,0		6,0
Modulgruppe Analysis, Numerics and Approximation Theory			15,0
Mathematics of Science	1,3 (a)		6,0
Introduction to One-parameter Semigroups of Operators with Background from Real and Functional Analysis	1,3 (a)		9,0
Modulgruppe Dynamical Systems and Optimization			18,0
Elements of the theory of solitons	1,3 (a)		6,0
Analysis of nonlinear dynamical systems	1,7 (a)		6,0
Modern theory of dynamical systems	1,3 (a)		6,0
Modulgruppe Stochastics, Statistics			6,0
Ergodic theory	3,3 (a)		6,0
Modulgruppe Data Analysis and Data Management and Programming			21,0
Principles of AI Engineering	2,0		6,0
Introduction to Deep Learning	3,7		6,0
Management of Scientific Data	2,7		6,0
What is Data Science?	2,3 (a)		3,0
Modulgruppe Key Competencies and Language Training			6,0
Occupational Internship	1,3 (a)		4,0
Anerkannte Schlüsselqualifikation (von ausländischer Hochschule)	1,0 (a)		2,0
Gesamtnote und Summe der ECTS-Leistungspunkte (LP)	1,8		121,0

Titel der Masterarbeit: Frequency Spectra

Berechnung der Gesamtnote gemäß der Allgemeinen Studien- und Prüfungsordnung für Masterstudiengänge vom 27. April 2016 in der derzeit gültigen Fassung (AStuPO) in Verbindung mit der Fachstudien- und -prüfungsordnung für den Studiengang „Computational Mathematics“ mit dem Abschluss Master of Science an der Universität Passau vom 27. November 2017 (FStuPO)

Für die Gesamtnote gilt: bis 1,1 (sehr gut, mit Auszeichnung), bis 1,5 (sehr gut), bis 2,5 (gut), bis 3,5 (befriedigend) und bis 4,0 (ausreichend)

Verwendete Abkürzungen in der Spalte „Bezeichnung“: Basismodul (BM), Grundkurs (GK), Hauptseminar (HS), Kolloquium (KO), Oberseminar (OS), Prüfungsleistung (PL), Praktikum (PR), Proseminar (PS), Praktikum (PT), Repetitorium (REP), Seminar (S), Seminar (SE), Tutorium (TU), Übung (Ü), Vorlesung (V), Vorlesung und Arbeitskurs (V/AK), Vorlesung (VL), Wissenschaftliche Übung (WÜ), Wissenschaftliche Übung für Fortgeschrittene (WÜF)

Annotationen in der Spalte „Anmerkungen“ („Anm.“): Anerkannte Leistungen von inländischer oder ausländischer Hochschule oder gleichwertige Kompetenzen als (a), die Note fließt nicht in die Berechnung der Gesamtnote ein als (b), die Leistung ist unbenotet als (c)

Diploma Supplement

This Diploma Supplement model was developed by the European Commission, Council of Europe and UNESCO/CEPES. The purpose of the supplement is to provide sufficient independent data to improve the international 'transparency' and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended.

1 INFORMATION IDENTIFYING THE HOLDER OF THE QUALIFICATION

1.1 Family name(s), First name(s)

Tregubov, Andrei

1.2 Date of birth (YYYY-MM-DD)

2001-03-31

1.3 Student identification code

117628

2 INFORMATION IDENTIFYING THE QUALIFICATION

2.1 Name of qualification and (if applicable) title conferred (in original language)

Master of Science (M.Sc.)

2.2 Main field(s) of study for the qualification

Computational Mathematics

2.3 Name and status of awarding institution (in original language)

Universität Passau, Fakultät für Informatik und Mathematik, University (state institution)

2.4 Name and status of institution (if different from 2.3) administering studies (in original language)

Not applicable.

2.5 Language(s) of instruction/examination

English

3 INFORMATION ON THE LEVEL AND DURATION OF THE QUALIFICATION

3.1 Level of the qualification

Level 7 of the European Qualifications Framework (EQF); Second Degree, with thesis; Master of Science.

3.2 Official duration of programme in credits and years

120 ECTS credits, 2 years.

3.3 Access requirement(s)

A first degree in Mathematics, Computer Science, Informatics or a related subject.

4 INFORMATION ON THE PROGRAMME COMPLETED AND THE RESULTS OBTAINED

4.1 Mode of study

Full-time.

4.2 Programme learning outcomes

Graduates of this Master's programme are able to thoroughly understand and independently work on challenging problems related to Computational Mathematics.

They are adept at precise formulation, conceptual and logical thinking as well as abstraction and discovering patterns and analogies. Proficient in recognising, formalising and analysing mathematical interrelationships in different fields, they are able to mathematically model highly complex theoretical and practical problems and to choose suitable mathematical methods to address and solve them.

Moreover, graduates acquire technical knowledge and team-working skills as well as the ability to present advanced mathematical notions both orally and in writing.

In addition to its theoretical foundation and orientation, this Master's programme has a strong emphasis on computational and application-related components with a wide range of modules focusing on mathematical software skills, algorithmic mathematics or applications such as cryptography, computer algebra, image processing and stochastic simulation.

Graduates also possess in-depth knowledge of applications of computational mathematics to other subjects such as data science, programming or economics and are therefore able to communicate and co-operate across disciplines.

The Master's programme prepares its graduates profoundly for their chosen field of profession, including independent work in the context of research and the advancement of scientific knowledge.

4.3 Programme details, individual credits gained and grades/marks obtained

See Final Transcript of Records.

4.4 Grading system and, if available, grade distribution table

Grading system for exams (passing grades only)

Grade descriptor	sehr gut	gut	befriedigend	ausreichend
Grades (exams)	1.0–1.3	1.7–2.3	2.7–3.3	3.7–4.0
Classification	very good	good	satisfactory	sufficient

Grading system for the degree

Grade descriptor	sehr gut, mit Auszeichnung	sehr gut	gut	befriedigend	ausreichend
Grades (degree)	1.0–1.1	1.2–1.5	1.6–2.5	2.6–3.5	3.6–4
Classification	very good, with distinction	very good	good	satisfactory	sufficient

Grade distribution table for the programme

52 candidates have successfully completed this degree since it was introduced.

Grade descriptor	Classification	Marking range	Distribution in cohort
sehr gut, mit Auszeichnung	very good, with distinction	1.0 – 1.1	25.00 %
sehr gut	very good	1.2 – 1.5	23.08 %
gut	good	1.6 – 2.5	48.08 %
befriedigend	satisfactory	2.6 – 3.5	3.85 %
ausreichend	sufficient	3.6 – 4.0	0.00 %

4.5 Overall classification of the qualification (in original language)

gut (1,8)

5 INFORMATION ON THE FUNCTION OF THE QUALIFICATION

5.1 Access to further study

Provides access to doctoral programmes (subject to the relevant institution's entry requirements with respect to grades and content).

5.2 Access to a regulated profession (if applicable)

Not applicable.

6 ADDITIONAL INFORMATION

6.1 Additional information

None.

6.2 Further information sources

None.

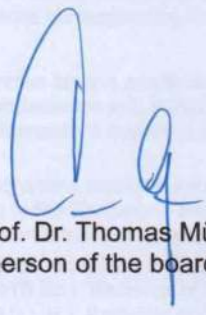
7 CERTIFICATION

This Diploma Supplement refers to the following original documents:

1. Urkunde (degree certificate), dated 1 April 2026
2. Zeugnis (examination certificate), dated 1 April 2026
3. Final Transcript of Records, dated 1 April 2026

Certification Date: 1 April 2026




Prof. Dr. Thomas Müller-Gronbach
Chairperson of the board of examiners

8. NATIONAL HIGHER EDUCATION SYSTEM¹

The information on the national higher education system on the following pages provides a context for the qualification and the type of higher education institution that awarded it.

8.1. Types of Institutions and Institutional Status

Higher education (HE) studies in Germany are offered at three types of Higher Education Institutions (HEI).²

- *Universitäten* (Universities) including various specialised institutions, offer the whole range of academic disciplines. In the German tradition, universities focus in particular on basic research so that advanced stages of study have mainly theoretical orientation and research-oriented components.
- *Fachhochschulen (FH)/Hochschulen für Angewandte Wissenschaften (HAW)* (Universities of Applied Sciences, UAS) concentrate their study programmes in engineering and other technical disciplines, business-related studies, social work, and design areas. The common mission of applied research and development implies an application-oriented focus of studies, which includes integrated and supervised work assignments in industry, enterprises or other relevant institutions.
- *Kunst- und Musikhochschulen* (Universities of Art/Music) offer studies for artistic careers in fine arts, performing arts and music; in such fields as directing, production, writing in theatre, film, and other media; and in a variety of design areas, architecture, media and communication.

Higher Education Institutions are either state or state-recognised institutions. In their operations, including the organisation of studies and the designation and award of degrees, they are both subject to higher education legislation.

8.2. Types of Programmes and Degrees Awarded

Studies in all three types of institutions have traditionally been offered in integrated "long" (one-tier) programmes leading to *Diplom- or Magister Artium* degrees or completed by a *Staatsprüfung* (State Examination).

Within the framework of the Bologna-Process one-tier study programmes are successively being replaced by a two-tier study system. Since 1998, two-tier degrees (Bachelor's and Master's) have been introduced in almost all study programmes. This change is designed to enlarge variety and flexibility for students in planning and pursuing educational objectives; it also enhances international compatibility of studies.

The German Qualifications Framework for Higher Education Qualifications (HQR)³ describes the qualification levels as well as the resulting qualifications and competences of the graduates. The three levels of the HQR correspond to the levels 6, 7 and 8 of the German Qualifications Framework for Lifelong Learning⁴ and the European Qualifications Framework for Lifelong Learning.⁵

1) The information covers only aspects directly relevant to purposes of the Diploma Supplement.

2) *Berufsakademien* are not considered as Higher Education Institutions, they only exist in some of the Länder. They offer educational programmes in close cooperation with private companies. Students receive a formal degree and carry out an apprenticeship at the company. Some *Berufsakademien* offer Bachelor courses which are recognised as an academic degree if they are accredited by the Accreditation Council.

3) German Qualifications Framework for Higher Education Degrees. (Resolution of the Standing Conference of the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany of 16 February 2017).

4) German Qualifications Framework for Lifelong Learning (DQR). Joint resolution of the Standing Conference of the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany, the German Federal Ministry of Education and Research, the German Conference of Economics Ministers and the German Federal Ministry of Economics and Technology (Resolution of the Standing Conference of the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany of 15 November 2012). More information at www.dqr.de.

5) Recommendation of the European Parliament and the European Council on the establishment of a European Qualifications Framework for Lifelong Learning of 23 April 2008 (2008/C 111/01 – European Qualifications Framework for Lifelong

Learning – EQF).

For details cf. sections 8.4.1, 8.4.2, and 8.4.3 respectively. Table 1 provides a synoptic summary.

8.3. Approval/Accreditation of Programmes and Degrees

To ensure quality and comparability of qualifications, the organisation of studies and general degree requirements have to conform to principles and regulations established by the Standing Conference of the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany (KMK).⁶ In 1999, a system of accreditation for Bachelor's and Master's programmes has become operational. All new programmes have to be accredited under this scheme; after a successful accreditation they receive the seal of the Accreditation Council.⁷

8.4. Organisation and Structure of Studies

The following programmes apply to all three types of institutions. Bachelor's and Master's study programmes may be studied consecutively, at various higher education institutions, at different types of higher education institutions and with phases of professional work between the first and the second qualification. The organisation of the study programmes makes use of modular components and of the European Credit Transfer and Accumulation System (ECTS) with 30 credits corresponding to one semester.

8.4.1. Bachelor

Bachelor's degree programmes lay the academic foundations, provide methodological competences and include skills related to the professional field. The Bachelor's degree is awarded after 3 to 4 years.

The Bachelor's degree programme includes a thesis requirement. Study programmes leading to the Bachelor's degree must be accredited according to the Interstate study accreditation treaty.⁸

First degree programmes (Bachelor) lead to Bachelor of Arts (B.A.), Bachelor of Science (B.Sc.), Bachelor of Engineering (B.Eng.), Bachelor of Laws (LL.B.), Bachelor of Fine Arts (B.F.A.), Bachelor of Music (B.Mus.) or Bachelor of Education (B.Ed.).

The Bachelor's degree corresponds to level 6 of the German Qualifications Framework/ European Qualifications Framework.

8.4.2. Master

Master is the second degree after another 1 to 2 years. Master's programmes may be differentiated by the profile types "practice-oriented" and "research-oriented". Higher Education Institutions define the profile.

The Master's degree programme includes a thesis requirement. Study programmes leading to the Master's degree must be accredited according to the Interstate study accreditation treaty.⁹

Second degree programmes (Master) lead to Master of Arts (M.A.), Master of Science (M.Sc.), Master of Engineering (M.Eng.), Master of Laws (L.L.M.), Master of Fine Arts (M.F.A.), Master of Music (M.Mus.) or Master of Education (M.Ed.). Master's programmes which are designed for continuing education may carry other designations (e.g. MBA).

The Master's degree corresponds to level 7 of the German Qualifications Framework/ European Qualifications Framework.

Learning – EQF).

6) Specimen decree pursuant to Article 4, paragraphs 1 – 4 of the interstate study accreditation treaty (Resolution of the Standing Conference of the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany of 7 December 2017).

7) Interstate Treaty on the organization of a joint accreditation system to ensure the quality of teaching and learning at German higher education institutions (Interstate study accreditation treaty) (Decision of the Standing Conference of the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany of 8 December 2016), Enacted on 1 January 2018.

8) See footnote 7.

9) See footnote 7.

8.4.3. Integrated "Long" Programmes (One-Tier): Diplom degrees, Magister Artium, Staatsprüfung

An integrated study programme is either mono-disciplinary (*Diplom* degrees, most programmes completed by a *Staatsprüfung*) or comprises a combination of either two major or one major and two minor fields (*Magister Artium*). The first stage (1.5 to 2 years) focuses on broad orientations and foundations of the field(s) of study. An Intermediate Examination (*Diplom-Vorprüfung* for *Diplom* degrees; *Zwischenprüfung* or credit requirements for the *Magister Artium*) is prerequisite to enter the second stage of advanced studies and specialisations. Degree requirements include submission of a thesis (up to 6 months duration) and comprehensive final written and oral examinations. Similar regulations apply to studies leading to a *Staatsprüfung*. The level of qualification is equivalent to the Master's level.

Integrated studies at *Universitäten* (U) last 4 to 5 years (*Diplom degree, Magister Artium*) or 3.5 to 6.5 years (*Staatsprüfung*). The *Diplom* degree is awarded in engineering disciplines, the natural sciences as well as economics and business. In the humanities, the corresponding degree is usually the *Magister Artium* (M.A.). In the social sciences, the practice varies as a matter of institutional traditions. Studies preparing for the legal, medical and pharmaceutical professions are completed by a *Staatsprüfung*. This applies also to studies preparing for teaching professions of some Länder. The three qualifications (*Diplom, Magister Artium* and *Staatsprüfung*) are academically equivalent and correspond to level 7 of the German Qualifications Framework/European Qualifications Framework. They qualify to apply for admission to doctoral studies. Further prerequisites for admission may be defined by the Higher Education Institution, cf. section 8.5.

Integrated studies at *Fachhochschulen* (FH)/*Hochschulen für Angewandte Wissenschaften* (HAW) (Universities of Applied Sciences, UAS) last 4 years and lead to a *Diplom* (FH) degree which corresponds to level 6 of the German Qualifications Framework/European Qualifications Framework. Qualified graduates of FH/HAW/UAS may apply for admission to doctoral studies at doctorate-granting institutions, cf. section 8.5.

Studies at *Kunst- und Musikhochschulen* (Universities of Art/Music etc.) are more diverse in their organisation, depending on the field and individual objectives. In addition to *Diplom/Magister* degrees, the integrated study programme awards include certificates and certified examinations for specialised areas and professional purposes.

8.5. Doctorate

Universities as well as specialised institutions of university standing, some of the FH/HAW/UAS and some Universities of Art/Music are doctorate-granting institutions. Formal prerequisite for admission to doctoral work is a qualified Master's degree (UAS and U), a *Magister* degree, a *Diplom*, a *Staatsprüfung*, or a foreign equivalent. Comparable degrees from universities of art and music can in exceptional cases (study programmes such as music theory, musicology, pedagogy of arts and music, media studies) also formally qualify for doctoral work. Particularly qualified holders of a Bachelor's degree or a *Diplom* (FH) degree may also be admitted to doctoral studies without acquisition of a further degree by means of a procedure to determine their aptitude. The universities respectively the doctorate-granting institutions regulate entry to a doctorate as well as the structure of the procedure to determine aptitude. Admission further requires the acceptance of the Dissertation research project by a professor as a supervisor.

The doctoral degree corresponds to level 8 of the German Qualifications Framework/ European Qualifications Framework.

8.6. Grading Scheme

The grading scheme in Germany usually comprises five levels (with numerical equivalents; intermediate grades may be given): "Sehr Gut" (1) = Very Good; "Gut" (2) = Good; "Befriedigend" (3) = Satisfactory; "Ausreichend" (4) = Sufficient; "Nicht ausreichend"

(5) = Non-Sufficient/Fail. The minimum passing grade is "Ausreichend" (4). Verbal designations of grades may vary in some cases and for doctoral degrees.

In addition, grade distribution tables as described in the ECTS Users' Guide are used to indicate the relative distribution of grades within a reference group.

8.7. Access to Higher Education

The General Higher Education Entrance Qualification (*Allgemeine Hochschulreife, Abitur*) after 12 to 13 years of schooling allows for admission to all higher educational studies. Specialised variants (*Fachgebundene Hochschulreife*) allow for admission at *Fachhochschulen* (FH)/*Hochschulen für Angewandte Wissenschaften* (HAW) (UAS), universities and equivalent higher education institutions, but only in particular disciplines. Access to study programmes at *Fachhochschulen* (FH)/*Hochschulen für Angewandte Wissenschaften* (HAW) (UAS) is also possible with a *Fachhochschulreife*, which can usually be acquired after 12 years of schooling. Admission to study programmes at Universities of Art/Music and comparable study programmes at other higher education institutions as well as admission to a study programme in sports may be based on other or additional evidence demonstrating individual aptitude.

Applicants with a qualification in vocational education and training but without a school-based higher education entrance qualification are entitled to a general higher education entrance qualification and thus to access to all study programmes, provided they have obtained advanced further training certificates in particular state-regulated vocational fields (e.g. *Meister/Meisterin im Handwerk, Industriemeister/in, Fachwirt/in* (IHK), *Betriebswirt/in* (IHK) und (HWK), *staatlich geprüfte/r Techniker/in, staatlich geprüfte/r Betriebswirt/in, staatlich geprüfte/r Gestalter/in, staatlich geprüfte/r Erzieher/in*). Vocationally qualified applicants can obtain a *Fachgebundene Hochschulreife* after completing a state-regulated vocational education of at least two years' duration plus professional practice of normally at least three years' duration, after having successfully passed an aptitude test at a higher education institution or other state institution; the aptitude test may be replaced by successfully completed trial studies of at least one year's duration.¹⁰

Higher Education Institutions may in certain cases apply additional admission procedures.

8.8. National Sources of Information

Kultusministerkonferenz (KMK) [Standing Conference of the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany]; Graurheindorfer Str. 157, D-53117 Bonn; Phone: +49[0]228/501-0; www.kmk.org; E-Mail: hochschulen@kmk.org

Central Office for Foreign Education (ZAB) as German NARIC; www.kmk.org; E-Mail: zab@kmk.org

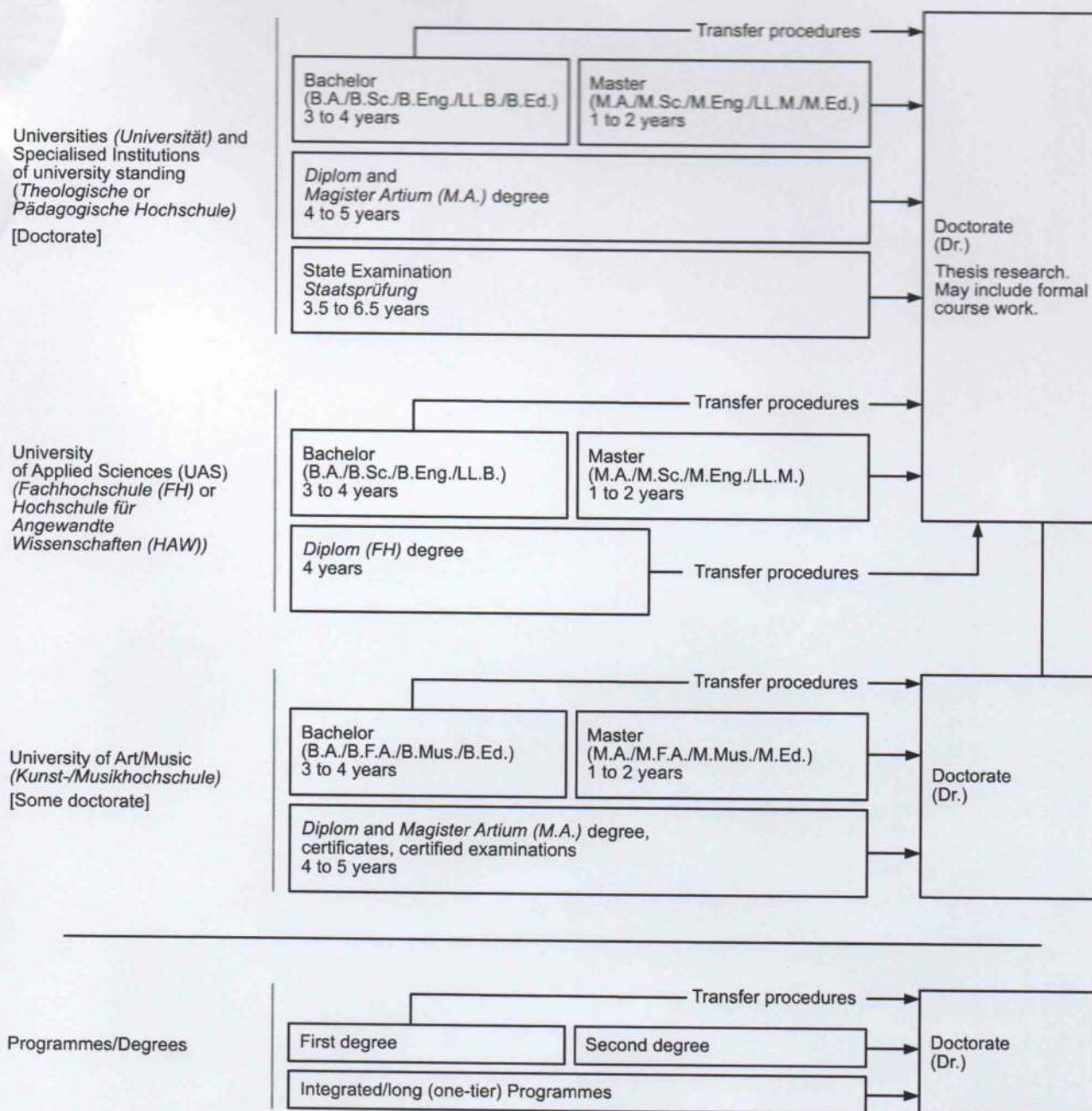
German information office of the Länder in the EURYDICE Network, providing the national dossier on the education system; www.kmk.org; E-Mail: Eurydice@kmk.org

Hochschulrektorenkonferenz (HRK) [German Rectors' Conference]; Leipziger Platz 11, D-10117 Berlin, Phone: +49 30 206292-11; www.hrk.de; E-Mail: post@hrk.de

"Higher Education Compass" of the German Rectors' Conference features comprehensive information on institutions, programmes of study, etc. (www.higher-education-compass.de)

¹⁰) Access to higher education for applicants with a vocational qualification, but without a school-based higher education entrance qualification (Resolution of the Standing Conference of the Ministers of Education and Cultural Affairs of the Länder in the Federal Republic of Germany of 6 March 2009).

Table 1: Types of Programmes and Degrees Awarded



Final Transcript of Records

Graduate

This is to certify that Andrei Tregubov, born on 31 March 2001 in Nischni Nowgorod, Russian Federation, has graduated from the degree programme "Computational Mathematics", Master of Science (M.Sc.).

Explanations

This document lists aggregate levels of the degree programme in boldface. Actual exams are listed in regular font weight. An aggregate level is any level from which the overall average mark and the sum of ECTS credits are calculated or is the level of the overall average mark itself. The module level, if existing, aggregates from the exam level directly below. There can be multiple exam levels below a given module level.

The date on the exam level is the date of the actual exam. The date on the module level is the exam date of the exam with the best grade, if applicable, or else the most recently passed exam. The date on other aggregate levels is the date of the most recently passed exam.

Abbreviations

advanced seminar (AS), colloquium (C), development practicum (EP), excursion (EX), foundation course (FC), lecture (L), lab practicum (P), practice time (PT), seminar (S), survey course (SC), tutorial (T), workshop (W), graduate seminar (GradSem), master class (MC), postgraduate seminar (PostSem), practice teaching (PracT), (ANK) recognised assessment - module level, (H) recognised assessment from the University of Passau - exam level, (D) recognised assessment from another higher education institution in Germany - exam level, (A) recognised assessment from another higher education institution outside of Germany - exam level, (SP ja) chosen focus area, if applicable, (ZEU) administrative annotation on graduation.

Grading system for exams and the degree (passing grades only)

Grade descriptor	sehr gut	gut	befriedigend	ausreichend
Grades (exams)	1.0–1.3	1.7–2.3	2.7–3.3	3.7–4.0
Classification	very good	good	satisfactory	sufficient

Grade descriptor	sehr gut, mit Auszeichnung	sehr gut	gut	befriedigend	ausreichend
Grades (degree)	1.0–1.1	1.2–1.5	1.6–2.5	2.6–3.5	3.6–4
Classification	very good, with distinction	very good	good	satisfactory	sufficient

Certification

This document is valid without signature.



Code	Title of module (unit)	Exam date	Grade	ECTS credits	Annotation
1000	Degree	2026-04-14	1.80	121	
9999	Accumulated credit/grade	2026-04-01	1.80	121	ZEU
481000	Compulsory modules	2026-04-01		40	
481001	Mathematics (S 1)	2025-03-25	2.30	5	
481001	Algorithms for Big Data	2025-03-25	2.30	5	
481002	Mathematics (S 2)	2026-04-01	1.00	5	ANK
481002	Mathematics (S 2)	2026-04-01	1.00	5	A
488999	Presentation M.Sc. Thesis	2026-04-01	1.00	3	ANK
488999	Presentation M.Sc. Thesis	2026-04-01	1.00	3	A
489900	M.Sc. Thesis	2026-04-01	1.30	27	ANK
489900	M.Sc. Thesis	2026-04-01	1.30	27	A
482000	Compulsory Elective Modules	2026-04-01		81	
482100	Module group: Algebra, Geometry and Cryptography	2025-08-01		15	
401816	Cryptography	2025-08-01	4.00	9	
401816	Cryptography	2025-08-01	4.00	9	
422150	Distributed Algorithms	2025-02-17	3.30	6	
422150	Distributed Algorithms	2025-02-17	3.30	6	
482300	Module group: Analysis, Numerics and Approximation Theory	2026-04-01		15	
455505	Mathematics of Science	2026-04-01	1.30	6	A
455505	Introduction to One-parameter Semigroups of Operators with Background from Real and Functional Analysis	2026-04-01	1.30	9	A
482400	Module group: Dynamical Systems and Optimization	2026-04-01		18	
455505	Elements of the theory of solitons	2026-04-01	1.30	6	A
455505	Analysis of nonlinear dynamical systems	2026-04-01	1.70	6	A
455505	Modern theory of dynamical systems	2026-04-01	1.30	6	A
482500	Module group: Stochastics, Statistic	2026-04-01		6	
455505	Ergodic theory	2026-04-01	3.30	6	A
482600	Module group: Data Analysis and Data Management and Programming	2026-04-01		21	
455410	Principles of AI Engineer	2025-02-12	2.00	6	

Code	Title of module (unit)	Exam date	Grade	ECTS credits	Annotation
455410	Principles of AI Engineering	2025-02-12	2.00	6	
471616	Introduction to Deep Learning	2025-02-19	3.70	6	
471616	Introduction to Deep Learning	2025-02-19	3.70	6	
472780	Management of Scientific Data	2025-02-21	2.70	6	
472780	Management of Scientific Data	2025-02-21	2.70	6	
455505	What is Data Science?	2026-04-01	2.30	3	A
482700	Module group: Applications				
482800	Module group: Key Competencies and Language Training	2026-04-01		6	
407680	Occupational Internship	2026-04-01	1.30	4	ANK
407680	Occupational Internship	2026-04-01	1.30	4	A
455506	Key Skills credits recognised (from study abroad)	2026-04-01	1.00	2	ANK
455506	Work in MMA "Plus"	2026-04-01	1.00	2	A